MIL-P-52310C(ME)
30 January 1984
SUPERSEDING
MIL-P-52310B
21 May 1975

MILITARY SPECIFICATION

PUMP, HYDRAULIC RAM, HAND-DRIVEN, 10,000 PSI

This specification is approved for use by the Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a hand-operated hydraulic pump used to supply pressure to various hydraulically operated tools.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4320

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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-H-52471/5	- Hose, Rubber, Hydraulic Pressure Type, Type 100R2 Double Wire Brand Reinforcement Hose.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The pump shall consist of a plunger-type pump, a pressure release valve, and an integral hydraulic-fluid reservoir. The pump shall be capable of being operated either vertically with pump head in down position or horizontally, and shall be equipped with all necessary fittings and high-pressure hoses.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.3.1 Material deterioration and control. The pump shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Pump. The pump shall be the single-cylinder, single-acting, plunger type, and shall develop not less than 10,000 pounds per square inch (psi) working pressure when using hydraulic fluid conforming to MIL-H-5606 or MIL-H-6083. The pump shall be equipped with an internal check valve to limit the loss of pressure to not more than 1000 psi during an 8-hour period after an initial "sit" time of 5 minutes. The pump shall be provided with a hand-operated pressure release valve, an integral hydraulic-fluid reservoir having a capacity of not less than 39 cubic inches (cu. in.), a filler opening fitted

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with a plug, and a handle provided with a grip to actuate the pump plunger. The plunger displacement shall be not less than 0.15 cu. in. ± 0.003 cu. in. A full pump stroke shall describe an arc of not more than 50 degrees with an effort of not more than 145 pounds. When tested as specified in 4.5.2.1, the pump, exclusive of the hydraulic-fluid reservoir, shall withstand a hydrostatic test pressure without leakage or permanent deformation. The pump shall weigh not more than 20 pounds, and it shall have mounting feet with holes or slots for mounting to any flat surface. The total length of the pump, with handle attached, shall be not more than 23 inches. The reservoir shall be furnished filled with hydraulic fluid conforming to MIL-H-5606 or MIL-H-6083.

3.5 Hose assembly. The pump shall be equipped with a high-pressure hose conforming to MIL-H-52471/5. The hose shall have an externally threaded 3/8 inch NPT corrosion-resistant or cadmium-plated steel fitting on one end and a corrosion-resistant quick-disconnect coupling on the other end. The hose assembly shall be provided with a complete coupling. The quick-disconnect coupling shall consist of both the male half-coupling and female half-coupling, furnished with a dust cap. The free end of the quick-disconnect coupling shall be externally threaded 3/8 inch NPT. When tested as specified in 4.5.2.1, the hose and quick-disconnect coupling shall withstand the hydraulic test pressure. Each end of the hose assembly shall be equipped with a wire spring guard not less than 3-3/4 inches long. The spring wire shall be corrosion-resistant or cadium-plated steel and shall be not less than 3/32 inch in diameter. The hose assembly shall be not less than 6 feet long.

3.5.1 Screw threads. Screw threads shall conform to FED-STD-H28.

3.6 Identification marking. The pump shall be identified in accordance with MIL-STD-130.

3.7 Treatment and painting. When specified (see 6.2), the portions of the pump normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.8 Workmanship. All parts, components, and assemblies of the hydraulic pump including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, and any other harmful extraneous material. All edges shall be rounded and chamfered.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article pump shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article pump shall be tested as specified in 4.5.2.1 through 4.5.2.6 Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 4.0 percent defective for major defects and 10.0 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Individual. Each pump shall be tested as specified in 4.5.2.1 and 4.5.2.2. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.3. AQL shall be 1.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The pumps shall be examined as specified herein for the following defects:

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Major

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatment.
105. Pump components not as specified.
106. Hose assembly not as specified.
107. Workmanship not as specified.

Minor

201. Pump reservoir not filled with hydraulic fluid.
202. Screw threads not as specified.
203. Identification marking missing, incomplete or not legible.
204. Treatment and painting not as specified.

4.5.2 Tests.

4.5.2.1 Hydrostatic test pressure. Subject the pump, hose and quick-disconnect couplings, exclusive of hydraulic fluid reservoir, to contractor's recommended test pressures for a period of not less than 2 minutes. Leakage, failure or permanent deformation of any component shall constitute failure of this test.

4.5.2.2 Working pressure. Fill the pump's hydraulic-fluid reservoir with the hydraulic fluid specified in 3.4 to the recommended level. Operate the pump in both the horizontal position and in the vertical position (with pump head in down position) to obtain specified working pressure. Inability of the pump to obtain and maintain the working pressure for a period of not less than 10 minutes shall constitute failure of this test.

4.5.2.3 Working-pressure loss. Operate the pump to obtain 10,000 psi working pressure. When the working pressure is obtained, cease operation of the pump and allow it to sit for a period of not less than 5 minutes. After the initial "sit" time, operate the pump to increase the working pressure back to 10,000 psi. Allow the pump to sit for a period of not less than 8 hours. A pressure loss of more than 1000 psi shall constitute failure of this test.

4.5.2.4 Handle effort. Operate the pump to obtain 10,000 psi working pressure and determine the maximum handle effort in pounds. A handle effort of more than 145 pounds shall constitute failure of this test.

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4.5.2.5 Fluid reservoir volume. Determine the volume of the hydraulic-fluid reservoir. Reservoir volume of less than 39 cu. in. shall constitute failure of this test.

4.5.2.6 Pump stroke. Operate the pump to determine full pump stroke in both horizontal and vertical positions. A full pump stroke in either position in which the handle describes an arc of more than 50 degrees shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of pack.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Materials or containers not as specified for level A or B. Each incorrect material or container shall constitute one defect.
109.	Pump not immobilized within the box as specified in level A.
110.	Box not closed and sealed by method specified for level A.
111.	Preservation and packing not in accordance with the referenced document as specified for commercial.
112.	Quantities packed together exceed the size or weight limitation of the box for level A or B.
113.	Marking missing, illegible, incorrect or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Unpainted exterior metal surfaces of the pump requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in, and shall be applied in accordance with, MIL-P-116. After the preservative has dried, if applied, each complete pump shall be placed in a close-fitting box conforming to PPP-B-636, W5c, style FOL. The pump shall be immobilized within the box to prevent movement or damage. Box closure and sealing shall be in accordance with method V of the appendix to the box specification.

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5.1.2 Commercial. Each complete pump shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. Pumps, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed the weight limitation of the box. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Pumps, preserved as specified in 5.1, shall be packed together as specified in 5.2.1, except the box shall be domestic type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13, or V15c, style optional or boxes conforming to PPP-B-640, class 2, style optional, may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Pumps, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The hydraulic pump is intended to be used as a pressure source for various hydraulically operated tools.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, date of this specification.

b. When a first article is required for inspection and approval, and numbers of pumps required (see 3.2).

c. When treatment and painting are required (see 3.7).

d. Degree of preservation and degree of packing required (see 5.1 and 5.2).

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6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction pump. The first article pump should consist of one or more pumps. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, tests, and approval of the first article.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and supplier's maintenance and operation manual to be furnished with each pump.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions regarding shipment of pump.

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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Review activity:
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User activities:
Army - AT, AV

Preparing activity:
Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-52310C(ME)	Pump, Hydraulic Ram, Hand-Driven, 10,000 PSI

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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